



UNIVERSITI TEKNIKAL MALAYSIA MELAKA

ASSIGNMENT 2

BERL3125 – SYSTEM INTEGRATION DESIGNING

CLO 3: Work individually or in groups effectively to perform assignments/tasks given (PLO 9, A3).

No	Name	Matrix No
1	MUHAMMAD AZRUL BIN REDZUAN	B122310626
2	NURSERI BINTI WASRI	B122310552
3	NUR FAQIHAH BINTI OTHMAN	B122310255

Week 9~15 Task: P&ID of industrial instrumentation system

This assignment should provide students with a practical opportunity to contribute to team effort by developing a P&ID.

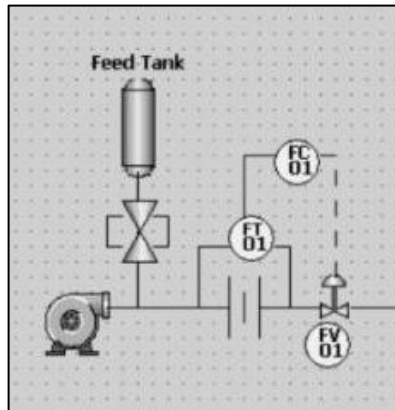
Objective: To analyze and develop a P&ID diagram from a PFD of a process by applying control loop techniques while observing related ISA guidelines.

In this assignment:

- 1) In your group, choose one industrial process with process flow diagram (PFD):
 - You can search the web for a PFD or use PFD examples in class lecture slide.
- 2) Develop a P&ID from the PFD:
 - You are free to add any sensors, control valves and controllers etc. as you deemed necessary
 - Expected in your P&ID: a minimum of 3 control loops
 - You can either draw your P&ID manually,
 - Or use any free online P&ID drawing software such as online.visual-paradigm.com and smartdraw.com.
- 3) In your report:
 - **Objective**
 - **Process Flow Diagram:** Show (cut and paste) the original PFD
 - **Piping and Instrumentation Diagram:** Show your developed P&ID
 - **Control Loop:** Describe the workings of each control loops in your P&ID.
 - This is a group assignment (max 3 students) – solo or one-person assignment group is not permitted.

OBJECTIVE

To convert a Process Flow Diagram (PFD) into a detailed Piping & Instrumentation Diagram (P&ID), identify equipment and unit operations, understand instrumentation and control loops, and demonstrate process knowledge for reporting and presentation.

PROCESS FLOW DIAGRAM

Flow Control

(MUHAMMAD AZRUL BIN REDZUAN B122310626)

COMPONENT:

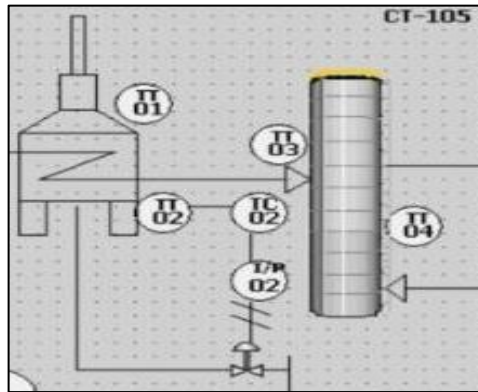
- Feed Tank
- Pump
- FT-01 (Flow Transmitter)
- FC-01 (Flow Controller)
- FV-01 (Flow Valve)

FUNCTION:

This section functions to supply raw materials into the system at a controlled flow rate to ensure a stable and consistent inlet flow before the heating and separation processes.

CONTROL LOOP #1:

The liquid from the feed tank is pumped into the system. FT-01 measures the actual flow rate and sends the signal to FC-01. FC-01 compares the measured value with the setpoint, and FV-01 opens or closes accordingly to control the flow rate.



Heat Exchanger & Temperature Control Section

(MUHAMMAD AZRUL BIN REDZUAN B122310626)

COMPONENT:

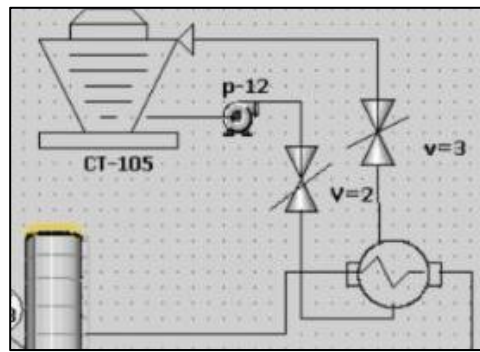
- Heat Exchanger
- TT-01, TT-02, TT-03, TT-04 (Temperature Transmitters)
- TC-02 (Temperature Controller)
- I/P-02
- Control Valve

FUNCTION:

This section is used to control the fluid temperature before it enters the column and to monitor the temperature at several critical locations to ensure process safety and product quality.

CONTROL LOOP #2:

The temperature transmitters (TT) measure the temperature at several points and transmit the data to TC-02. TC-02 controls the control valve via an I/P converter to adjust the flow of the heating or cooling medium in order to achieve the desired temperature.



Cyclone / CT-105 Section

(NURSERI BINTI WASRI B122310552)**COMPONENT:**

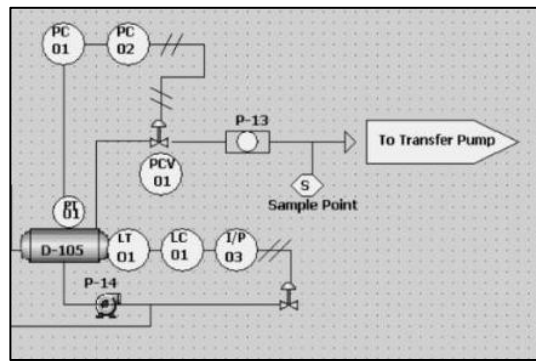
- CT-105 (Cyclone Separator)
- P-12 (Pump)
- Valve V-2 & V-3

FUNCTION:

It is used to separate solid particles or heavier materials from the main flow. In the P&ID, the valves and pumps are shown in detail.

CONTROL LOOP #3:

- The flow enters CT-105, the solids settle to the bottom, and the clarified flow exits to the next stage of the system.



Condenser & Drum D-105 Section

(NUSERI BINTI WASRI B122310552)

COMPONENT:

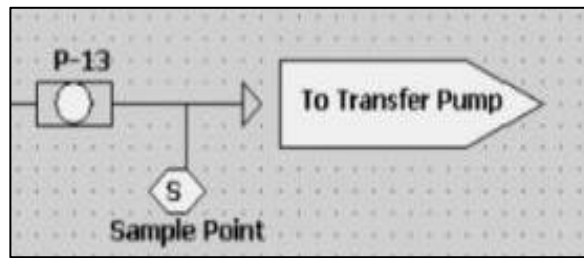
- Condenser
- D-105 (Drum / Separator)
- LT-01 (Level Transmitter)
- LC-01 (Level Controller)
- PC-01 / PC-02 (Pressure Controller)
- PT-01 (Pressure Transmitter)

FUNCTION:

This section cools the vapor and collects the product in the drum. It demonstrates a complete closed-loop control system (sensor → controller → actuator).

CONTROL LOOP #4:

- The vapor is cooled in the condenser, and the liquid collects in D-105. LT-01 measures the liquid level, LC-01 controls pump P-14, and the pressure is regulated by PC-01/PC-02 for safety.
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Product Transfer & Sampling Section

(NUR FAQIHAH BINTI OTHMAN B122310255)

COMPONENT:

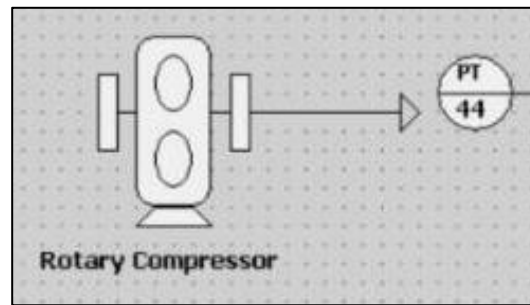
- Pump P-13
- Sample Point
- Line ke Transfer Pump

FUNCTION:

It is used to transfer the final product to the next unit and to take samples.

CONTROL LOOP #5:

The product is pumped out by P-13, samples can be taken for quality control (QC), and the product is transferred to the next system.



Rotary Compressor & Gas Line Section

(NUR FAQIAH BINTI OTHMAN B122310255)

COMPONENT:

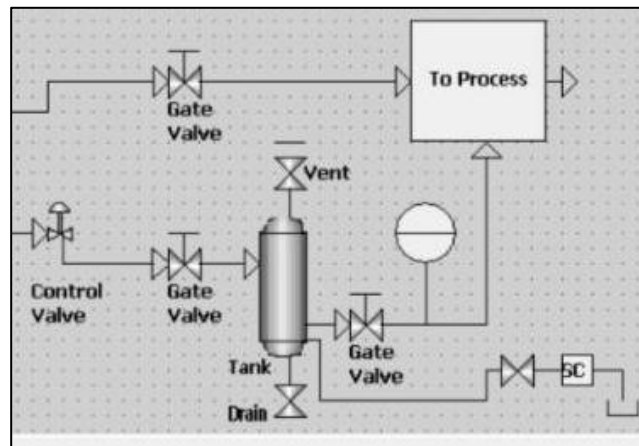
- Rotary Compressor
- PT-44 (Pressure Transmitter)

FUNCTION:

It provides gas or support pressure to the process.

CONTROL LOOP #6:

The compressor increases the gas pressure, and PT-44 monitors the pressure.



Tank, Vent & Drain Section

(NUR FAQIHAH BINTI OTHMAN B122310255)

COMPONENT:

- Tank
- Vent
- Drain
- Gate Valve
- Control Valve
- Pressure Transmitter

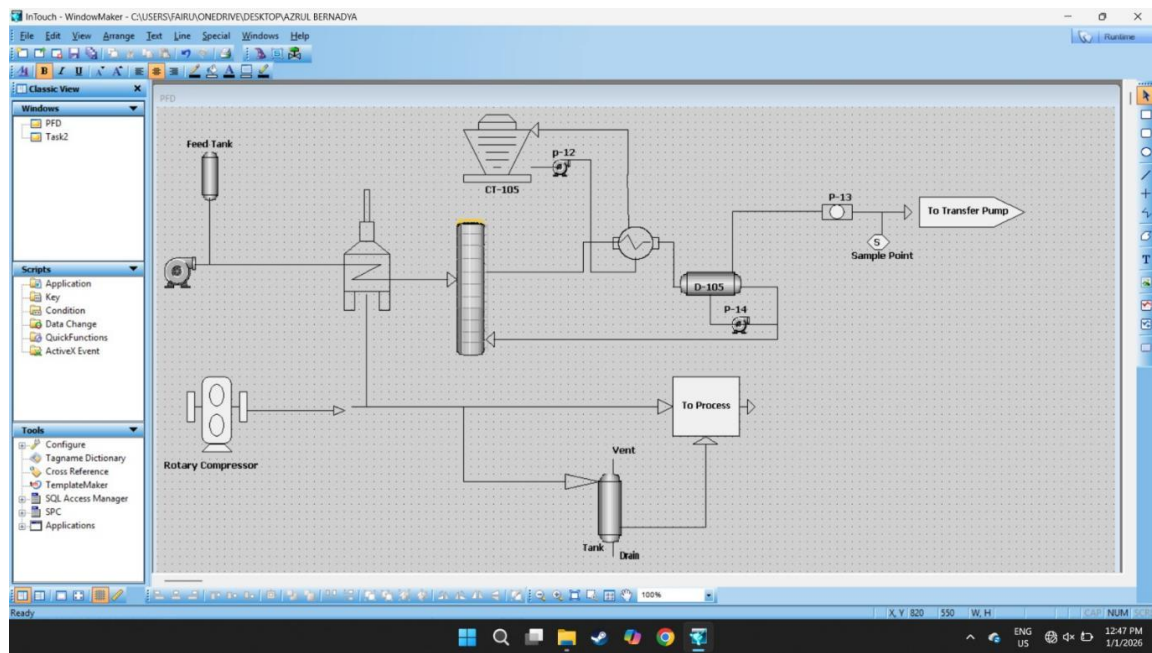
FUNCTION:

- Pressure release (vent)
- Liquid disposal (drain)
- Maintenance operations

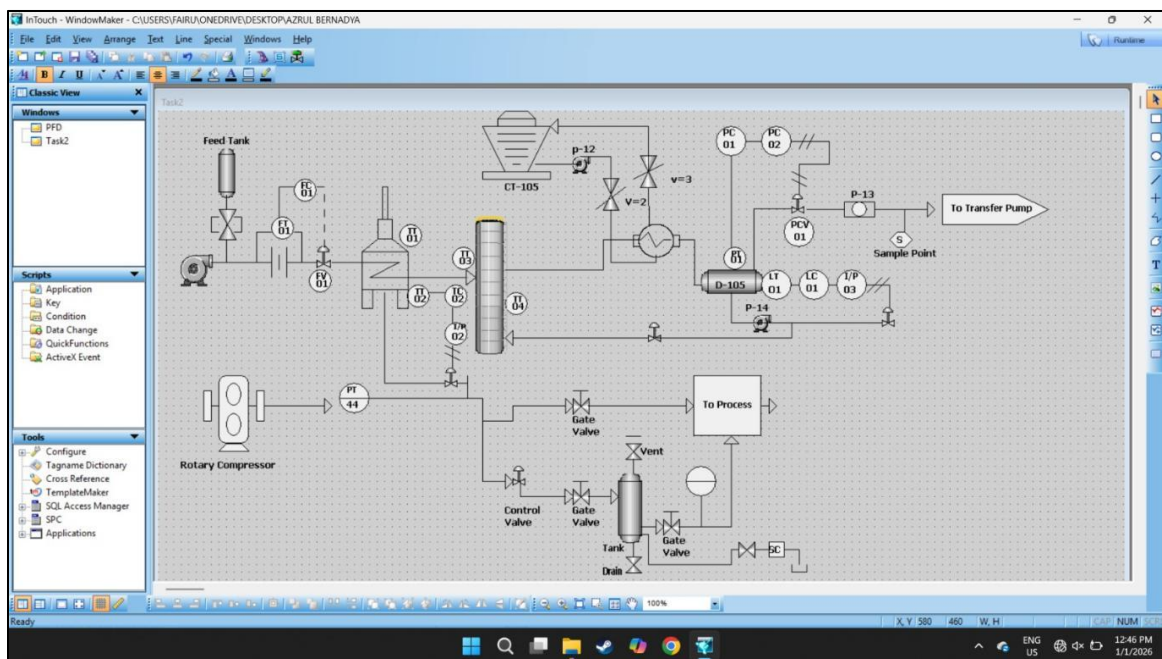
CONTROL LOOP #7:

Tank, Vent & Drain Section is used to control liquid level, ensure pressure safety, and allow maintenance operations. The tank level is monitored by a level transmitter and controlled by a level controller to keep the liquid within a safe operating range. The vent line acts as a pressure relief system to release excess pressure and prevent over-pressurization, while the drain line is manually operated for maintenance or emergency purposes and does not form part of an automatic control loop.

Full PFD Diagram:



Full P&ID Diagram:



“PFD only shows the major process flow, while P&ID provides detailed information on instrumentation, control loops, valves, safety devices and piping required for operation and maintenance.”

DETAILS FOR ASSIGNMENT ASSESSMENT RUBRICS

No	Items	Fully Met (4)	Met (3)	Partially Met (2)	Not Met (1)	Wght	Student 1	Student 2	Student 3	Student 4
Shared Group Marks										
1	Objective and Process Flow Diagram	Excellent	Good	Adequate	Poor	1.0				
2	Piping and Instrumentation Diagram	Excellent	Good	Adequate	Poor	2.0				
3	Control Loops	Excellent	Good	Adequate	Poor	2.0				
Individual Marks										
3	Quantity of Contribution (Is the student's contribution significant to the group assignment)	Significant	Good	Adequate	Poor	2.5				
4	Quality of Contribution (Is the student's contribution technically well-written)	Excellent	Good	Adequate	Poor	2.5				
5	Format, Grammar, Clarity and Organization	Excellent	Good	Adequate	Poor	2.5				

*Mark = Weight X Score

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